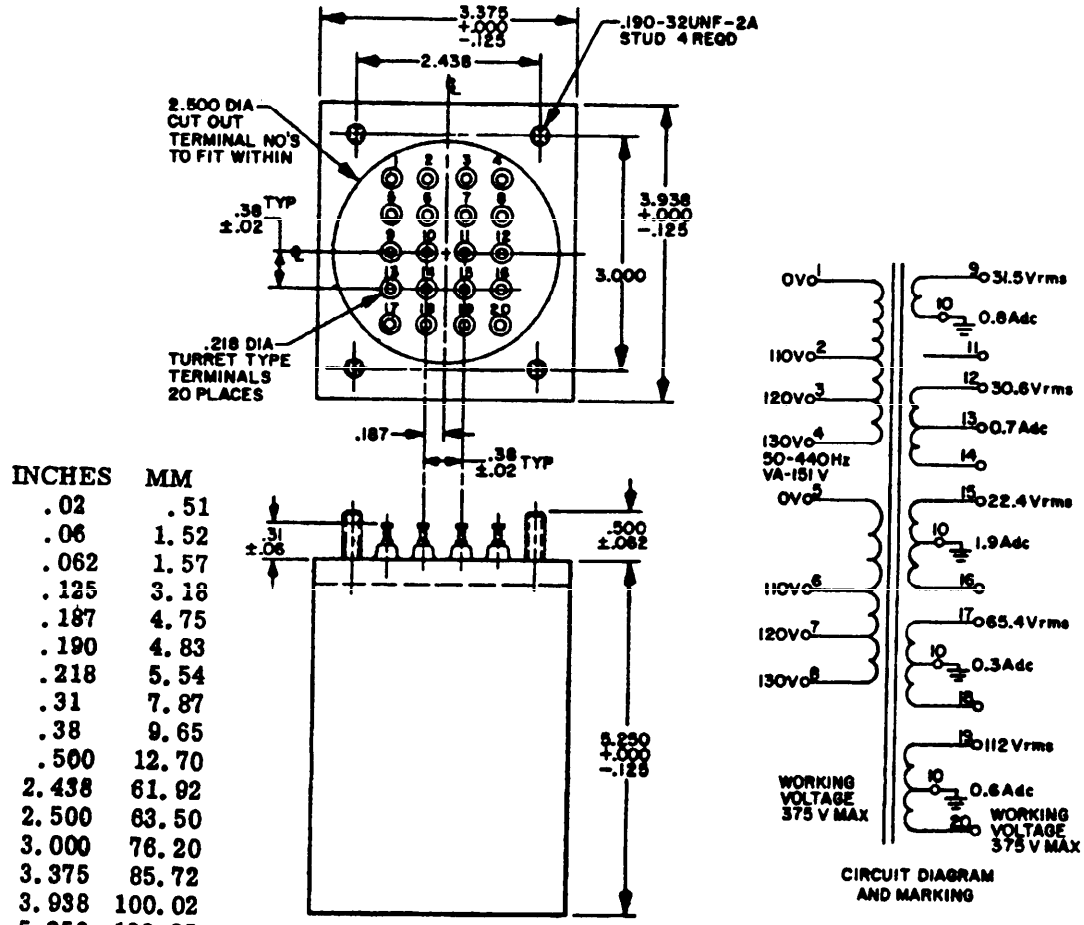
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-DOWN, TF4R03KA921

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.



INCHES	MM
.02	.51
.06	1.52
.062	1.57
.125	3.18
.187	4.75
.190	4.83
.218	5.54
.31	7.87
.38	9.65
.500	12.70
2.438	61.92
2.500	63.50
3.000	76.20
3.375	85.72
3.938	100.02
5.250	133.35

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.010 (.25 mm).
4. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configuration.

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PSC 5950

REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:
- Primary voltage:
  - (1-3) (5-7): 120 volts rms ±10, 50 Hz to 440 Hz.
- Secondary voltage:
  - (9-11): 31.5 volts rms, 0.80 Adc.
  - (12-14): 30.6 volts rms, 0.7 Adc.
  - (15-16): 22.4 volts rms, 1.90 Adc.
  - (17-18): 65.4 volts rms, 0.3 Adc.
  - (19-20): 112 volts rms, 0.60 Adc.
- DC resistance:
  - (1-4): 4.5 ohms maximum.
  - (5-8): 5.0 ohms maximum.
  - (5-6)(9-11): 0.47 ohm maximum.
  - (12-14): 0.68 ohm maximum.
  - (15-16): 0.14 ohm maximum.
  - (17-18): 2.07 ohms maximum.
  - (19-20): 3.62 ohms maximum.
- Working voltage:
  - (1-8): 375 volts maximum.
  - (9-20): 375 volts maximum.

Design and construction:
- Dimensions and configuration: See figure 1.
- Duty cycle: Continuous.
- Case: Metal.
  - Material: Steel.
- Terminals: Solder, turret type.
  - Terminal height: 0.310 inch ±0.060.
- Weight: 10.0 pounds.
- Operating temperature range: -55°C to +85°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 5.0 pounds.

Dielectric withstanding voltage:
- At sea level - 1050 volts rms.
- At barometric pressure - 945 volts rms.

Electrical characteristics:
- No load: With 120 volts rms, 60 Hz across (1-3) (5-7) in parallel, current in (1-3) shall not exceed 175 milliamperes and the power in (1-3) shall not exceed 4.8 watts.
  - Voltage across (9-11): 33.0 volts ct, ±5 percent.
  - Voltage across (12-14): 32.0 volts ct, ±5 percent.
  - Voltage across (15-16): 24.0 volts ct, ±5 percent.
  - Voltage across (17-18): 70.0 volts ct, ±5 percent.
  - Voltage across (19-20): 115 volts ct, ±5 percent.
- Rated load: With 120 volts rms, 60 Hz across (1-3) (5-7) in parallel:
  - Voltage across (9-11): 31.5 volts rms, ±5 percent.
  - Voltage across (12-14): 30.6 volts rms, ±5 percent.
  - Voltage across (15-16): 22.4 volts rms, ±5 percent.
  - Voltage across (17-18): 65.4 volts rms.
  - Voltage across (19-20): 112 volts rms.

Temperature rise: +28.73°C with 120 volts rms, 60 Hz across (1-3) at an ambient temperature of +65°C.

Marking location: See figure 1.
- Part number: M27/102-01.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, SG
Navy - EC, SH, OS
Air Force - 11, 17, 99
DSA - ES

User activities:
Army - MU, WC, SG, ME
Navy - MC, AS, SH, OS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5950-0512-2)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-27/102 TRANSFORMER, POWER , STEP-DOWN, TF4RO3KA921

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. *IS THE SPECIFICATION RESTRICTIVE?*

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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